

FPGA-accelerated fiber optic position detection

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ABSTRACT

To address the processing latency of high-resolution fiber positioning algorithms in large-scale spectral telescope systems, an FPGA-based acceleration architecture is designed. A spatial-domain parallel processing architecture is proposed, utilizing the MCDMA IP core for the segmentation and distribution of image data. By implementing parallel processing across multiple image slices, the architecture effectively resolves the bandwidth mismatch between data transmission and processing. The spot recognition and localization algorithm is optimized into a lightweight version by stripping the complex label assignment mechanisms and equivalence table merging modules inherent in traditional Run-Length Encoding (RLE). This reduces BRAM and logic resource consumption and eliminates global data dependencies, providing sufficient resource margins for massive parallelization. Simulation results demonstrate that multi-channel parallel processing significantly enhances throughput, achieving a frame rate of 20 FPS for 7920×6004 resolution images.

Keywords: FPGA, MCDMA, RLE, high-resolution images

1. INTRODUCTION

Observational facilities serve as the core engines driving astronomical breakthroughs. By capturing multi-wavelength data through ground-based and space-based observatories, scientists are able to conduct in-depth analyses of the evolutionary patterns of stars and galaxies. To satisfy the increasing demands for spectroscopic observation, numerous projects have emerged and played a transformative role in the field, such as LAMOST,¹ SDSS,² DESI,³ and 4MOST.⁴ Contemporary astronomy is advancing toward larger apertures, wider fields of view, and higher resolution, necessitating the continuous upgrading of astronomical facilities, as exemplified by China's recently implemented JUST project.⁵ While most initiatives adopt camera-feedback-based closed-loop control for fiber positioning, the processing latency of existing algorithms when handling high-resolution images and massive fiber units has become a primary bottleneck, hindering both precise control and observational efficiency. For instance, the LAMOST fiber positioning system achieves the precise alignment of 4,000 fibers within ten minutes, with an average image processing time of approximately 10 seconds.^{1,6}

For fiber positioning motion control, most projects implement feedback loops by utilizing camera metrology systems to acquire focal plane images and measure the coordinates of backlit fibers. Anti-collision motion planning is then performed based on this positional data, followed by the actuation control of the fiber positioning units, as exemplified by LAMOST.⁷ Future astronomical endeavors are trending toward the miniaturization and high-density integration of these units. When combined with the shift toward high-resolution observation, these developments pose severe challenges to the speed of fiber positioning, necessitating more advanced algorithms to optimize the process and enhance the overall operational efficiency of survey telescopes.

This research focuses on optimizing the backlit fiber position extraction algorithm within the context of real-time closed-loop control. As astronomical observational requirements escalate, systems are increasingly adopting

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industrial cameras with resolutions of 50 megapixels (50MP) or higher. The primary challenge in processing such ultra-large-scale image data lies in the significant processing latency. Traditional serial processing architectures based on CPUs or GPUs struggle to complete the concurrent extraction and centroid calculation of thousands of spots within a millisecond-level feedback cycle. Furthermore, the inherent complexity and suboptimal real-time performance of existing spot recognition and localization algorithms constitute a critical technical bottleneck, directly limiting the improvement of the closed-loop control frequency in fiber positioning systems.

To address these limitations, this research adopts Field Programmable Gate Arrays (FPGAs) as the algorithmic platform, owing to their native parallelism and deterministic latency. As programmable hardware systems, FPGAs provide high-concurrency processing capabilities and ultra-low signal latency; consequently, numerous research teams have leveraged FPGAs to achieve high-performance image processing in recent years. For instance, Qiyun Shen et al. utilized edge detection algorithms on FPGAs to extract spot centroids from LAMOST high-resolution images.⁸ However, this approach did not fundamentally resolve the challenges posed by high resolution, and the processing time per frame remained substantial. Similarly, Guifeng Chen employed FPGAs for spot centroid extraction. While this method achieved certain performance gains over previous work, the reliance on pipelined processing for massive high-resolution datasets continues to impose significant throughput constraints.⁹

This study leverages the inherent parallelism of FPGAs to design an acceleration scheme for real-time fiber recognition and localization. By implementing a high-performance parallel architecture for spot extraction on an FPGA hardware acceleration platform, the system achieves high-speed processing of image data from fiber positioning units. Simulation experiments demonstrate a pipelined throughput of 20 frames per second (FPS) for images with a resolution of 7920×6004 . Validation through centroid calculation shows a 100% recognition rate compared to PC-based results, with accuracy fully satisfying project requirements. This scheme is well-suited for integration into future fiber positioning systems.

2. SYSTEM ACCELERATION SCHEME ARCHITECTURE DESIGN

The primary objective of this research is to accelerate spot recognition and localization for real-time fiber positioning within the LAMOST system when integrated with high-resolution industrial cameras. However, the ultra-high resolution of these cameras creates a significant disparity where data transmission bandwidth far exceeds processing capacity. In handling massive images with tens of millions of pixels (e.g., 7920×6004), conventional "simple parallelization" or "basic pipelining" architectures often encounter physical hardware limitations, failing to meet project requirements. Specifically, processing a single pixel per clock cycle at a 100 MHz frequency results in a per-frame processing time exceeding 500 ms, which is incompatible with the camera's output of 8–10 FPS. While increasing the master clock frequency could theoretically improve throughput, the inherent physical properties and architectural constraints of FPGAs impose a definitive ceiling. Overclocking beyond these limits leads to compilation failures in timing closure, exponential increases in power consumption, and risks of hardware degradation.

In typical FPGA-based image processing workflows, image data is generally buffered in DDR4 memory and continuously streamed to functional modules via the AXI4 bus. However, when handling high-resolution images featuring tens of millions of pixels, the logic frequency of a single-channel processing pipeline often fails to match the GB/s-level bandwidth of the AXI4 bus. As shown in Figure 1, the combination of extreme resolution and dense spot distribution causes traditional serial processing modes to become a significant throughput bottleneck. Consequently, the algorithms must be redesigned and customized for the target imagery to fully exploit the hardware parallelism and pipelined architecture inherent in FPGAs.

To this end, this research proposes an FPGA-based real-time spot acceleration algorithm specifically designed for high-resolution images featuring multiple target spots. The overall system architecture is depicted in Figure 2.

This scheme leverages the high-efficiency access characteristics of FPGAs to fixed memory addresses by incorporating a Multi-Channel Direct Memory Access (MCDMA) module to construct a spatial-domain slice parallel processing architecture. As illustrated in Figure 3, the multi-channel mechanism of the MCDMA partitions the contiguous ultra-large image data stored in DDR4 into multiple parallel spatial slice data streams. The

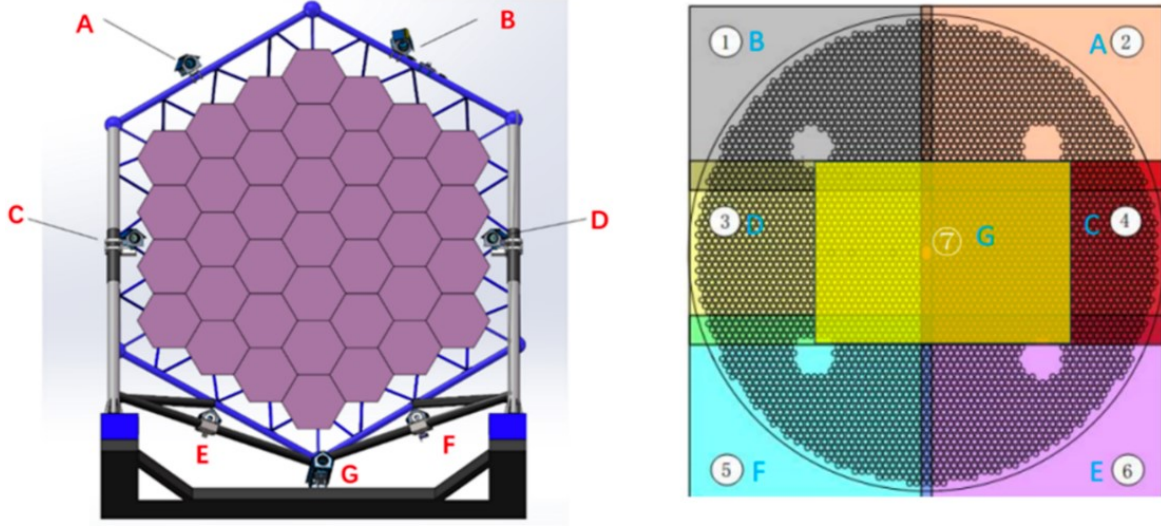


Figure 1: The mounting position of the fiber optic inspection camera around the primary mirror MB (left) and the distribution of the area covered by the corresponding camera on the focal plane (right).

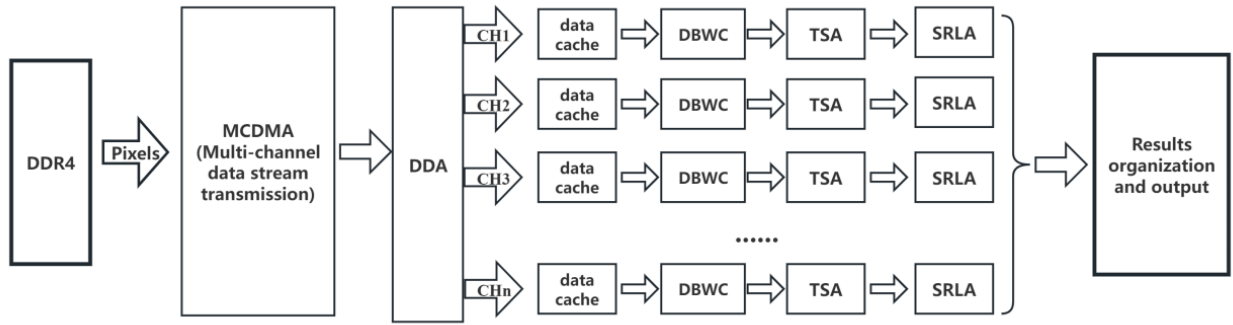


Figure 2: System Architecture (DDA:Data Distribution Algorithm. DBWC:Data Bit Width Conversion. TSA:Threshold Segmentation Algorithm. SRLA:Spot Recognition and Localization Algorithm).

starting address and data volume for each stream are determined based on the image's location in DDR4 and the designated number of slices. The MCDMA tags these distinct data streams for round-robin transmission, allowing subsequent algorithms to distribute data according to their respective stream identifiers. This strategy disperses the ultra-high bandwidth requirement of a single channel across multiple parallel processing engines, each equipped with an independent one-pass processing pipeline. By adjusting the number of channels, the system can flexibly match the total processing bandwidth of the backend algorithms with the transfer width of the frontend AXI4 bus, effectively resolving the bandwidth mismatch between high-speed data transmission and slower algorithmic processing.

Furthermore, to ensure that the backend parallel processing units are not constrained by FPGA hardware resources, this study performs a comprehensive optimization of the spot recognition and localization algorithm. By decoupling the complex label assignment mechanisms and equivalence table merging modules inherent in traditional Run-Length Encoding (RLE), a lightweight algorithm was developed to address on-chip resource constraints while maximizing throughput. This approach significantly reduces the consumption of BRAM and logic resources and eliminates global data dependencies, thereby providing essential resource margins for the spatial-domain parallelization of high-resolution images. The design overcomes the scalability bottleneck previously associated with RLE-based spot recognition on FPGAs, achieving an optimal balance between processing

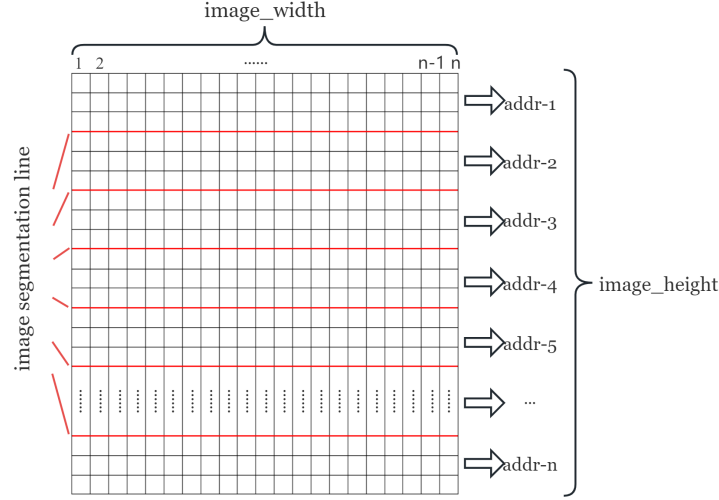


Figure 3: Image data segmentation diagram.

performance and hardware utilization.

3. PARALLELIZED ARCHITECTURE FOR SPOT RECOGNITION AND LOCALIZATION ALGORITHM

3.1 Multi-channel data distribution

Prior to processing, image data is typically buffered in DDR4 memory. In the hardware implementation, ultra-high-resolution raw data is initially written to the DDR4 storage via a high-speed DMA engine. This strategy facilitates bandwidth decoupling between the sensor interface and the algorithmic logic—ensuring the algorithm remains independent of specific sensor data formats and transmission rates—while establishing the data foundation required for subsequent spatial-domain parallelization.

Within the Xilinx Vivado environment, this architecture utilizes the AXI-based MCDMA (Multi-Channel Direct Memory Access) IP core. Unlike traditional AXI DMA IP cores, which are limited to a single read channel (MM2S) and a single write channel (S2MM) per instance, the MCDMA supports up to 16 independent channels within a single IP instance. Implementing an image-partitioning scheme with 8 or 10 parallel slices using standard AXI DMA would require the instantiation of 8 or 10 separate IP cores, leading to excessive overhead in control logic and registers. In contrast, the MCDMA shares underlying AXI data bus interfaces and scheduling logic across multiple channels, thereby preserving significant hardware resources for the implementation of parallel centroid calculation units.

As illustrated in Figure 4, two storage regions are established within the DDR4 memory: one for raw image data and the other for the Scatter-Gather (SG) linked list required by the MCDMA’s SG mode. Each entry in the SG linked list represents a physically contiguous block of DDR4 storage. The DMA controller retrieves data from the DDR4 sequentially based on the information provided in the SG entries. The Descriptor serves as the primary logical unit of an SG transfer, consisting of the Buffer Address (the start of a contiguous storage space), the Next Descriptor Pointer (the address of the following descriptor), Bytes to Transfer (the data volume managed by the descriptor), and the Descriptor Control (for memory data mapping). Each transmission channel is mapped to a specific descriptor chain. Once the CPU updates the linked list and initiates the MCDMA, the IP core interacts with the SG chains via the SG bus to output image data from specific memory locations. Through custom configurations, the system can switch between chains after transmitting a specific data packet. By

assigning weights to the transmission channels, the data volume per channel can be controlled; in this design, equal weights are used to facilitate sequential transmission according to channel IDs. For instance, in an 8-channel configuration, the system transmits a data packet from Channel 0, proceeds sequentially through the other channels, and resumes with Channel 0 after Channel 7 has completed its transfer.

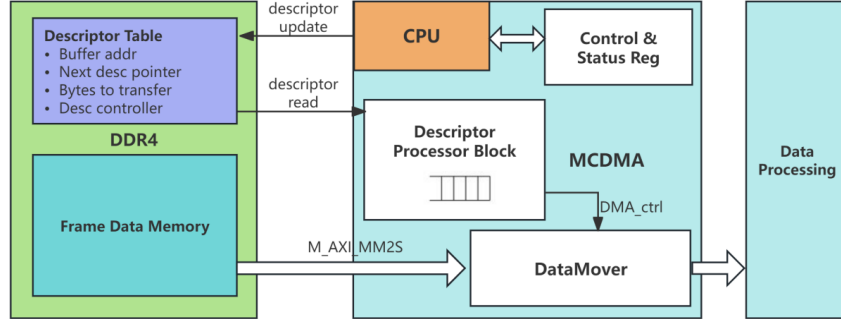


Figure 4: Image data segmentation diagram.

An efficient data distribution and buffering mechanism is designed for the multi-channel data streams output by the MCDMA. The core logic of this module involves identifying the TDEST signal within the aggregated AXI4-Stream data, which allows the multiplexed stream to be demultiplexed into corresponding parallel processing channels based on predefined destination identifiers. By implementing the spatial-domain slice parallelization strategy, the processing rate of the backend algorithms is dynamically matched with the frontend transmission bandwidth. Owing to this throughput symmetry, the system utilizes a lightweight configuration for the FIFO buffers in each channel; the FIFO depth is required to be only slightly larger than the data packet size of a single-channel transfer. This design ensures continuous data flow while significantly minimizing the consumption of FPGA BRAM (Block RAM) resources.

To maximize transmission efficiency, DDR4 and the AXI bus typically utilize a physical bit-width of 128 bits or larger, whereas the backend spot extraction algorithm primarily performs logic operations on single pixels (e.g., 8-bit grayscale values). Within this architecture, the FIFO serves as a spatial buffer to facilitate the disassembly of high-bandwidth data into 16 consecutive 8-bit pixel units. Concurrently, in the temporal domain, a "wide-write, narrow-read" flow regulation mechanism is implemented to smooth intermittent burst data into a continuous and stable low-bit-width pixel stream with a constant step size. This conversion mechanism effectively resolves timing synchronization issues between different processing layers, ensuring that the backend pipeline maintains deterministic timing for high-efficiency processing even during full-speed transmission of high-resolution images.

Considering the stable space environment and the high uniformity in the luminance distribution of captured images, alongside the adoption of a spatial-domain slice parallel processing architecture, this study omits complex preprocessing algorithms such as global histogram equalization or sophisticated adaptive filtering. Such algorithms frequently rely on full-frame or neighborhood statistical characteristics, which can lead to image feature distortion or discontinuities at slice boundaries during partitioned processing, thereby increasing the complexity and error rate of subsequent cross-region spot merging. To mitigate random noise, the system integrates suppression logic directly into the spot recognition algorithm instead of utilizing an independent filtering stage. By applying pixel area thresholds and geometric constraints during the extraction of connected domain features, real-time noise rejection is realized. This strategy further streamlines the hardware pipeline and reduces the total resource overhead of the system.

3.2 Optimization of Spot Recognition Algorithm

This research addresses the challenge of spot recognition in images exceeding 50 megapixels, requiring the precise identification of more than 1,000 backlit fibers. Given the rigorous demands for real-time performance

and accuracy, a highly optimized Run-Length Encoding (RLE) algorithm is proposed. By streamlining the algorithmic logic, this approach achieves high processing throughput with minimal hardware resource overhead, thereby effectively satisfying the stringent requirements for massive spatial-domain parallelization.

The overall algorithm is primarily divided into two functional stages: the Parallelized Spot Recognition and Localization Module and the Spot Merging and Output Management Module, as illustrated in the figure 5 below.

The first stage consists of multiple logically symmetrical sub-modules that directly interface with the image slice data streams distributed by the MCDMA. Utilizing a “one-pass scanning” strategy, these sub-modules synchronously extract the fundamental geometric features of spots as the pixel stream flows through. The resulting feature information is then passed to the second stage, which is responsible for the logical classification and cross-region reorganization of the aggregated feature streams. Specifically, this stage identifies and classifies spots that were truncated at the boundaries during the initial segmentation, temporarily storing them in an on-chip feature buffer to await matching data from adjacent slices. Ultimately, the system reconstructs and outputs the data for the complete spots.

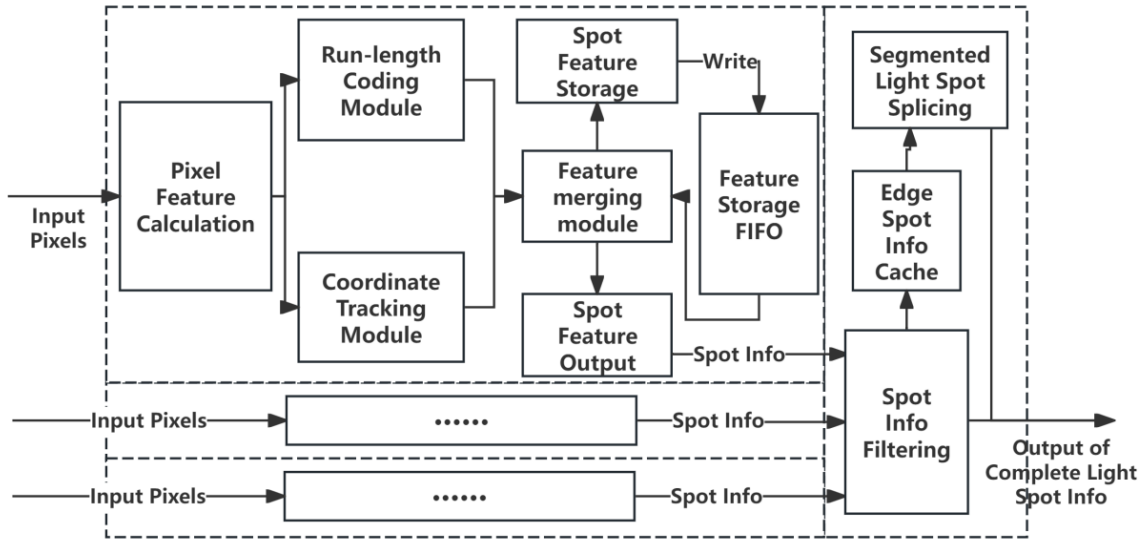


Figure 5: Light spot recognition algorithm structure diagram.

3.2.1 Pixel Gray Level Feature Calculation

Considering the limited FPGA resources and stringent real-time requirements, the grayscale centroid method is prioritized for spot extraction, with the calculation principle provided in Equation 1 :

$$x_{centroid} = \frac{\sum I_i x_i}{\sum I_i}, \quad y_{centroid} = \frac{\sum I_i y_i}{\sum I_i}, \quad (1)$$

By identifying connected components and calculating their intensity-weighted centers, the system achieves high-speed, low-overhead sub-pixel localization. Compared to computationally intensive edge-fitting or deep-learning algorithms, the grayscale centroid method offers inherent linear pipelining characteristics.

During this processing stage, the system synchronously calculates the grayscale accumulation and the weighted coordinate sums within each run-length unit as the pixel stream is scanned. By integrating dedicated multipliers to compute coordinate weights in real-time during Run-Length Encoding (RLE) and utilizing a highly synchronized pipelined architecture, the system performs instantaneous data accumulation and clock-synchronized output. This approach not only ensures strict timing continuity and data stream compactness but also significantly minimizes FPGA resource consumption through logic reuse and parallel design.

3.2.2 Image Tagging Based on Run Length Coding

Similar to traditional Run-Length Encoding (RLE), this algorithm performs connected component determination upon pixel input and outputs the corresponding feature information. However, unlike conventional RLE methods, the feature information in this approach is updated on a row-by-row basis. The feature storage space is optimized to maintain only the data for a single row; once processing for the current row is completed, the storage is iteratively updated for the subsequent row, rather than storing the entire frame for batch processing.

Consequently, the feature structure defined in this algorithm differs from traditional formats. As shown in Equation 2 and 3, the feature vector variable is defined as:

$$\text{Feature} = \{begin, length, label, x_{weight}, y_{weight}, I_{sum}, count\}, \quad (2)$$

$$\begin{cases} begin = x \\ length = 1 \\ label = 0 \end{cases} \quad \begin{cases} x_{weight} = \text{sum}(x * I) \\ y_{weight} = \text{sum}(y * I) \\ I_{sum} = \text{sum}(I) \\ count = \text{sum}(length) \end{cases} \quad (3)$$

During the row-wise encoding process, since pixels have already undergone threshold segmentation, a run unit's starting point is identified when a non-zero pixel follows a zero pixel. At this point, the feature parameters are initialized:

- *begin* and *length*: Represent the standard RLE parameters, denoting the starting horizontal coordinate and the unit length, respectively.
- *x_{weight}*, *y_{weight}*, and *I_{sum}*: Represent the x-coordinate weighted sum, y-coordinate weighted sum, and the cumulative grayscale intensity.
- *label*: A specialized parameter used to indicate whether a component is located at the image boundary to facilitate the subsequent merging of truncated spots. Values of 0 and 1 represent internal and boundary spots, respectively.
- *count*: Records the number of pixels within the component to enable real-time noise suppression based on area thresholds.

This study focuses primarily on the spot recognition of backlit fibers. Although the target spots are predominantly regular, irregularities persist across the thousands of detected units. Specifically, local morphology gaps at the apex or base of a spot frequently induce a misidentification phenomenon, as illustrated in Figure 6.

If a conventional Run-Length Encoding (RLE) algorithm were employed, these irregularities would be incorrectly identified as multiple independent units. Such misidentification severely disrupts the subsequent feature iteration logic and can lead to the loss of critical information during the feature merging process with the succeeding row. Furthermore, while specialized global merging algorithms could be designed to handle these edge cases, such complexity would significantly inflate the burden on on-chip feature storage and increase the consumption of logic resources, contradicting the requirement for a lightweight architecture.

Using the highly regular morphology of the spot imagery, an enhanced Run-Length Encoding (RLE) algorithm is developed to ensure that each spot feature within a single row is represented by a single set of refined vectors. This approach optimizes the data stream and simplifies subsequent iterative update logic. The core mechanism of the algorithm involves real-time continuity determination for short-distance gaps: when the gap width between two feature sets in a row is less than 2 pixels and the effective pixel lengths on both sides exceed 2 pixels, the system automatically integrates them into a single feature entity. Conversely, if these conditions are not met, the data is purified by discarding short, redundant segments. This strategy preserves critical pixel information while removing noise, effectively balancing algorithmic complexity with hardware resource utilization. As illustrated in Figure 6, since the left portion of the last row is less than 2 pixels, it is discarded to maintain the integrity of the core feature.

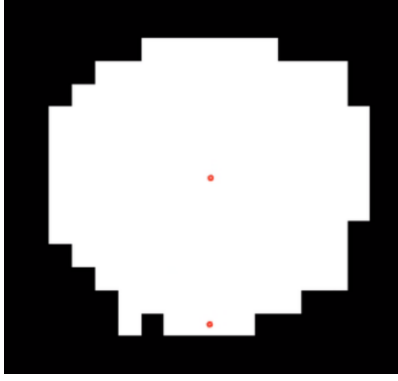


Figure 6: An irregular light spot was calculated to have two centroids.

3.2.3 Spot Feature Mosaic Based on FIFO Characteristics

Following the row-wise run-length encoding (RLE) of the image, each scanning line generates a series of run-length feature vectors. To mitigate the substantial on-chip resource overhead associated with multi-row feature buffering in conventional architectures, this research proposes a real-time merging architecture based on single-row feature rolling backtracking. This architecture utilizes a FIFO module to buffer features for only a single row. By leveraging the inherent first-in-first-out streaming characteristics of the FIFO, the system performs real-time retrieval and comparison of feature states from the preceding row while simultaneously processing the run-lengths of the current row.

The core of this merging mechanism is the dynamic matching logic of the coordinate tracking module. When the coordinate information indicates a spatial intersection between the current run and a feature from the preceding row, the system executes online aggregation and updates the feature vector. If the current run finds no matching counterpart in its upper neighborhood, it is stored in the FIFO as a newly identified connected component. Conversely, if a feature retrieved from the FIFO representing the preceding row has no valid connectivity along the current scanning path, the spot is determined to have reached physical closure, which immediately triggers the feature output pipeline for centroid calculation.

As illustrated in Figure 7, the coordinate tracking module provides the coordinates of the pixel currently being processed. The pixels in the red box represent the rows of pixels that are being processed. The operational sequence is as follows:

1. **FIFO Retrieval:** When the scan reaches position ①, the system triggers the FIFO to output the feature information from the preceding row, which is then temporarily stored in a buffer register to facilitate subsequent comparisons.
2. **Connectivity Detection:** Position ② corresponds to the termination coordinate of a connected component in the preceding row. During the interval between positions ① and ②, the system monitors for valid pixels. If no connectivity is detected within this range, the corresponding feature is deemed closed and is output directly.
3. **Feature Update and Storage:** Upon detecting the pixel at position ③, identifying the end of a connected component, and confirming it meets the area threshold (filtering noise), the system directs the FIFO to store the data. If a merger was identified during step (2), the integrated feature vector is stored; otherwise, it is recorded as an independent, new connected component.

During the feature merging stage, the system updates the geometric properties of a spot by inheriting the starting position (*begin*), length, and label from the current row's run-length unit. Simultaneously, real-time incremental operations are performed on the remaining accumulated variables, such as intensity sums and weighted

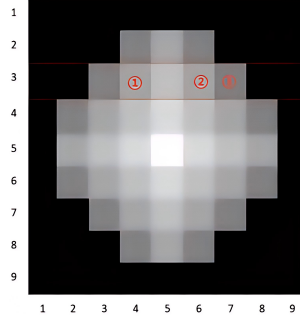


Figure 7: The algorithm steps are explained through simulated light spots.

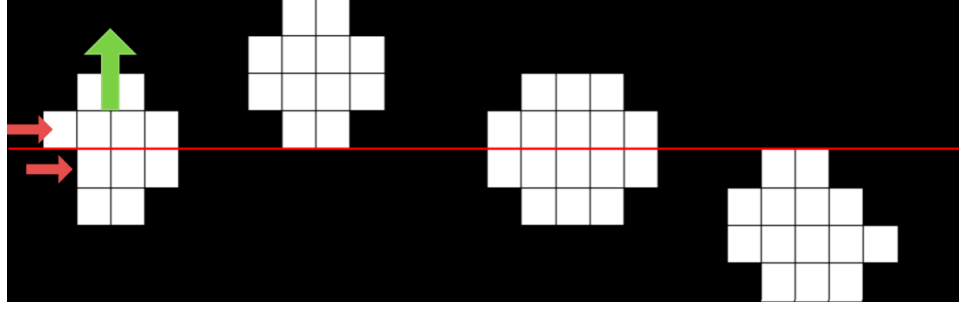
coordinates. This architecture employs a FIFO-based rolling buffer mechanism to store the features of a single row. Because the number of run-length units per row is orders of magnitude smaller than the raw pixel count, the memory footprint is drastically reduced compared to frame-buffer-based approaches. This optimized design, which consumes only a single FIFO IP core and minimal logic resources, significantly alleviates the FPGA hardware burden. This efficiency is the architectural cornerstone that enables the real-time processing of ultra-high-resolution imagery within strict hardware constraints.

3.2.4 Feature Information Organization and Output of Light Spot Characteristics

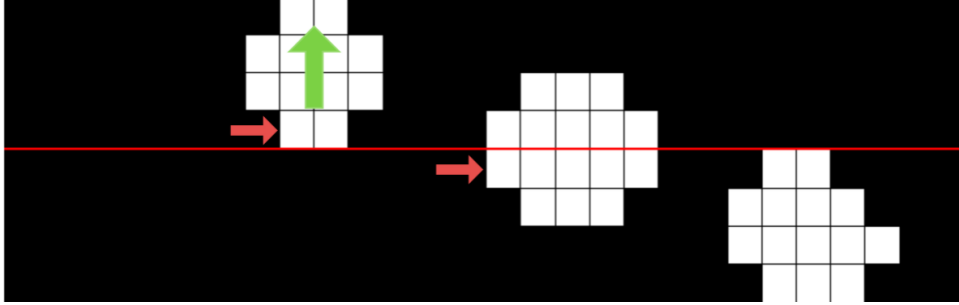
Due to the adoption of a parallelized slice processing architecture, spots spanning the partition boundaries are split into two independent feature sets. Failure to reorganize these fragmented features would lead to misjudgment of the final localization and a significant reduction in sub-pixel accuracy. To address this fragmentation, the system incorporates a cross-region feature integration strategy. Feature information tagged with a boundary collision identifier ($label = 1$) is stored in a dedicated buffer based on its starting coordinate (*begin*). Once all parallel channels have completed their respective processing tasks, a logic unit performs topological stitching of these fragmented features based on their spatial neighborhood relationships, as shown in the figure 8a and 8c. For fragmented segments where no matching counterpart is retrieved (as shown in Figure 8b and 8d), the system outputs the optical spot located at the front as a fully closed optical spot. This ensures the integrity and consistency of spot feature extraction across the entire massive parallel processing environment.

To manage the asynchronous feature streams output by the various parallel processing channels, the system incorporates a multi-stage buffering and transmission architecture where a lightweight primary FIFO buffer unit is implemented at each channel's exit. This design is specifically engineered to absorb instantaneous congestion that occurs when multiple channels output feature information simultaneously, effectively preventing data collisions and bottlenecks at the convergence point. Following initial arbitration, the feature data from all paths is integrated into a deeper secondary aggregation FIFO. Once the data volume within this buffer reaches a pre-defined burst transmission threshold, the system invokes the AXI DataMover IP core to initiate high-efficiency burst write operations, transferring the accumulated feature data in batches to designated result regions within the DDR4 memory.

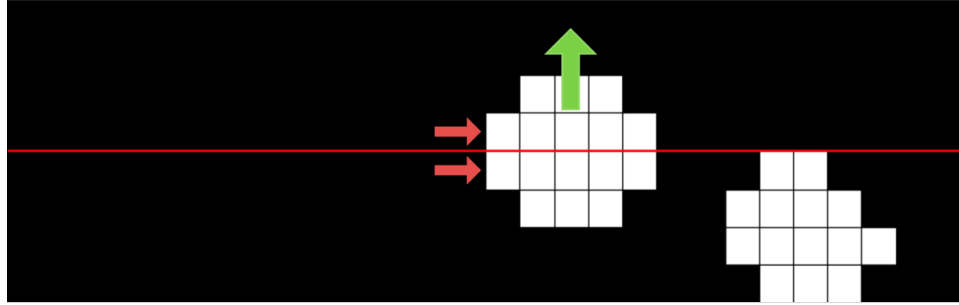
Furthermore, to guarantee real-time performance for images with tens of millions of pixels, the feature storage link utilizes an independent AXI bus channel that provides physical isolation from the raw image data stream. This "dual-bus shunt" strategy eliminates bandwidth competition between result output and image acquisition, which effectively reduces the risk of bus congestion. By segregating these high-bandwidth data flows, the architecture significantly enhances bus utilization and the overall data throughput efficiency of the system.



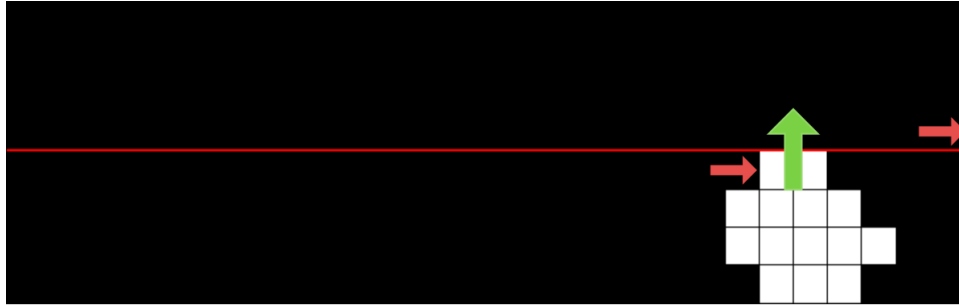
(a) The segmented light spots are stitched together, and the feature information from the upper and lower groups is merged to generate light spot information.



(b) The second light spot merely coincides with the edge on the cutting line without splicing, and only the upper feature information is processed.



(c) The third segmented light spots are stitched together, and the feature information from the upper and lower groups is merged to generate light spot information.



(d) The fourth light spot merely coincides with the edge on the cutting line without splicing, and only the below feature information is processed

Figure 8: Schematic diagram of edge-cutting light spot combined with

4. EXPERIMENTAL CONCLUSIONS AND RESULT ANALYSIS

To validate the acceleration performance of the proposed algorithm for spot recognition and localization in high-resolution images, relevant simulation experiments were designed. Utilizing observation images from the LAMOST project, an analysis of the hardware resource utilization, accuracy, and real-time performance of the proposed algorithm was conducted. The hardware validation platform selected for this study is the Alinx XCZU4EV development board, featuring the XCZU4EV-1SFVC784I FPGA model.

4.1 Hardware resource analysis

The architectural framework of the proposed algorithm consists of three primary components: MCDMA data distribution, parallelized spot recognition, and the consolidated output of encoding results. The consumption of memory resources is primarily concentrated in the MCDMA multi-channel FIFO buffers used to interface with bus transmissions. The capacity of these buffers is strictly calibrated to match the data bit-width and single-burst length while maintaining a reasonable engineering margin.

In contrast, the feature storage FIFOs and the RAM used for processing fragmented spots within the parallel recognition units are characterized by a highly lightweight design. These units only require buffering of the run-length encoding information for a single row. By integrating multiple cumulative variables—such as area and coordinate weights—into a single vector, the bit-width for each feature vector was determined through simulation to be 100 bits. Considering the sparse distribution of spots in 50-megapixel images and the presence of noise interference, a storage depth of 128 was established to prevent data overflow while ensuring sufficient operational headroom. The remaining segments of the pipeline utilize only minimal-capacity auxiliary FIFOs to maintain timing synchronization, thereby maximizing the optimization of FPGA hardware resource allocation while ensuring the real-time processing of high-resolution imagery.

A resource consumption comparison was conducted between a single module of the proposed algorithm and an HLS-designed algorithm from Reference .⁹ As shown in the table 1 below, the proposed architecture achieves a substantial reduction in hardware resource utilization compared to the HLS implementation. By minimizing the reliance on complex high-level abstractions and optimizing the data path at the RTL level, the design significantly lowers the consumption of logic and memory resources. This extreme efficiency provides the necessary physical conditions for the massive parallel deployment of the algorithm, enabling the integration of multiple processing channels within the limited fabric of the FPGA.

Table 1: FPGA resource consumption comparison

RLE algorithm	LUTs	FF	BRAM	DSP
algorithm from Chen ⁹	7796	6408	12	10
Current algorithm	1078	1098	3	2

4.2 Analysis of Localization Accuracy and Real-Time Performance

To validate the acceleration performance of the proposed algorithm for high-resolution spot recognition and localization, a hardware-software co-simulation experimental platform was designed and constructed. Due to hardware environment constraints, the experiment pre-stores multiple high-resolution raw images in DDR4 memory and drives the MCDMA to execute cyclic read operations to simulate the continuous input of high-bandwidth video streams. By deploying a frame-end interrupt mechanism within the processing architecture, the system triggers a serial port identifier upon the completion of each frame. Timestamps received by the host computer are then used to precisely calculate system throughput, thereby verifying the real-time response performance under multi-megapixel loads. Additionally, the spot feature packets output by the FPGA hardware were extracted and processed in MATLAB for centroid calculation and accuracy comparison. Through dispersion analysis of the theoretical models and the hardware calculation results, the sub-pixel localization accuracy of the system was comprehensively evaluated.

The acceleration effects of the algorithm are presented in Table 2. As shown, the processing time of the algorithm deployed on the FPGA is significantly faster than its execution on a PC. The performance on the FPGA is directly correlated with the number of image slice channels, n ; a higher number of partitioned channels results in increased processing speeds. When configured with 10 channels, the frame rate reaches 20 FPS, and the processing bandwidth achieves 7 Gbps, successfully meeting the project’s requirements.

Table 2: The impact of the number of image slice channels on processing latency

Algorithm deployment	PC	FPGA(n=1)	FPGA(n=5)	FPGA(n=8)	FPGA(n=10)
Single-frame processing time	965ms	263ms	74.9ms	55.6ms	47.1ms

To assess the precision of the proposed architecture, the spot feature information output by the FPGA was compared against results generated by the grayscale centroid method and the ellipse fitting method executed on a PC. This comparison serves as the basis for its accuracy assessment. The simulation results using field-captured images are shown in Figure 9, where it is evident that every spot is accurately identified and localized. The localization results from the FPGA were benchmarked against MATLAB computations, with the resulting error distributions shown in Figure 10. Furthermore, an analysis of the dispersion for each localized point was conducted to compare the FPGA’s performance with PC-based methods, as summarized in the table 3 below.

Experimental data indicates that while slight deviations occur in spots with low pixel counts (smaller spots), the remaining target points are localized with high precision. This confirms that the hardware-based implementation maintains sub-pixel accuracy consistent with theoretical models while operating at significantly higher throughput.

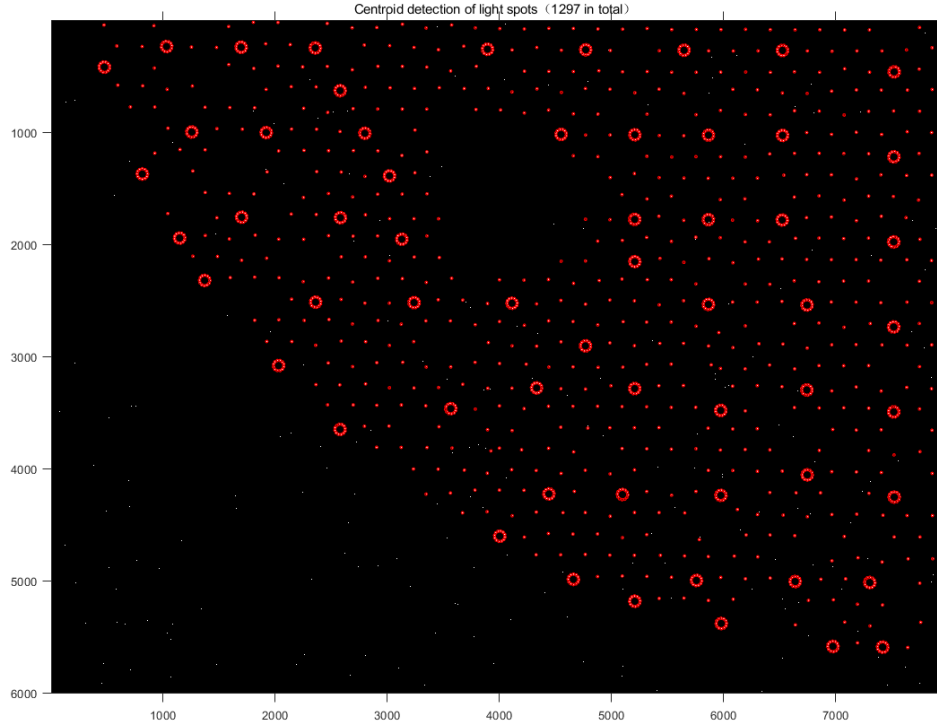


Figure 9: Computation results deployed on FPGA.

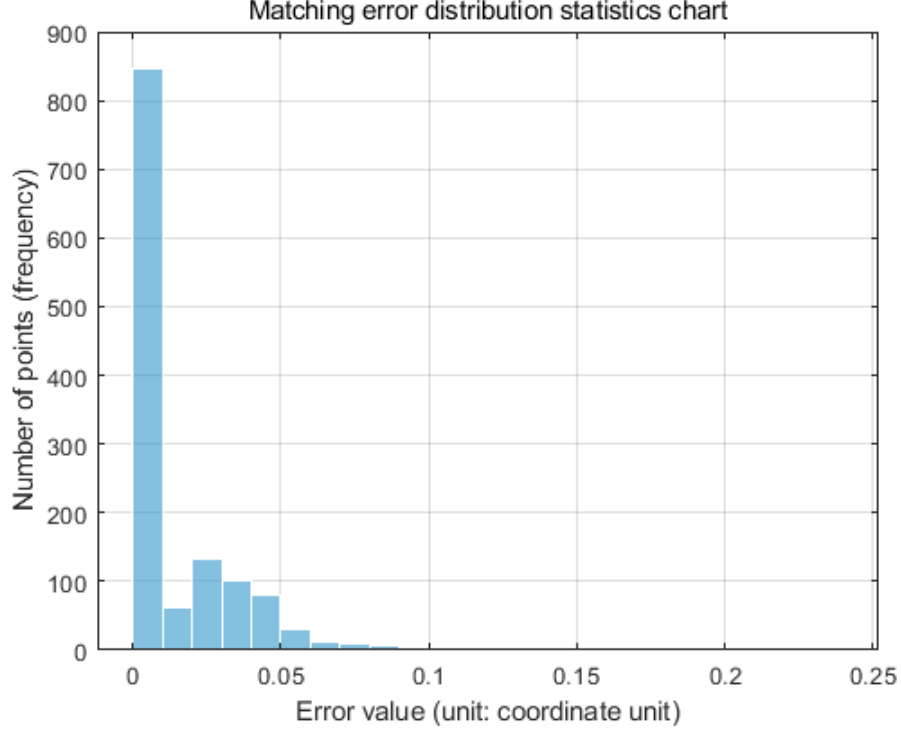


Figure 10: Error statistical histogram.

Table 3: A comparative error analysis was conducted in MATLAB using several different methodologies to evaluate the algorithm’s precision.

	Variance	Standard Deviation	Maximum Error
Weighted Centroid Method	0.000552	0.023501	0.233326
Ellipse Fitting Method	0.002063	0.045424	0.308937

5. CONCLUSION

This study addresses the urgent requirement for real-time processing of ultra-high-resolution imagery in fiber positioning systems by designing and implementing a high-performance, FPGA-based parallel spot extraction architecture. By utilizing the MCDMA IP core in conjunction with a Scatter-Gather descriptor linked-list mechanism, high-bandwidth transmission of raw high-resolution image data between the DDR4 memory and the processing cores was successfully achieved. Furthermore, the implementation of a spatial-domain slicing parallelization design effectively overcame the physical bandwidth bottlenecks inherent in single-channel processing.

A key innovation of this work is the proposed enhanced Run-Length Encoding (RLE) algorithm. By introducing "automatic gap connection" and "single-row feature backtracking iteration" mechanisms, the algorithm transforms complex connected component labeling into a pipeline operation with extremely low resource overhead. Experimental results demonstrate that the design fulfills project requirements using only a single FIFO IP core and minimal logic resources, significantly reducing BRAM utilization and allowing the system to support a higher degree of parallel expansion.

Hardware-software co-simulation experiments confirm that the architecture meets the closed-loop feedback requirements of the LAMOST system in terms of both processing rate and accuracy when handling 50-megapixel images. The research presented here provides a resource-efficient and highly reliable engineering paradigm for

real-time information extraction from ultra-large-scale image sensors, offering significant practical value. This solution also establishes a robust foundation for future system upgrades to meet the challenges of even larger-scale observations and higher frame rates.

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